Exercise 6-2: Clean the Cars data

Did you use a language model?

A : No, I did not find any use for a language model for this assignment except for writing comments.

Read the data

In [1]:
```
# Imports the Pandas library
import pandas as pd
```

In [2]:
```
# Load the car data from a CSV file into a DataFrame.
cars = pd.read_csv('cars.csv')
```

In [3]:
```
# Display the first few rows of the 'cars' DataFrame to get an initial understan
cars.head()
```

Examine the data

In [4]:
```
# Show a concise summary of the 'cars' DataFrame, including data types and non-n
cars.info()
```

In [5]:
```
# Display the counts of unique values in the 'fueltype' column to understand the
cars.fueltype.value_counts()
```

In [6]:
```
# List all unique car names to identify the range and variety of cars included i
cars.CarName.unique()
```

Fix spelling and capitalization problems in the data

In [7]:
```
# Create a new column 'brand' by extracting the first word from 'CarName', assum
cars['brand'] = cars.apply(lambda x: x.CarName.split(' ')[0], axis=1)
# Create a new column 'name' by joining the remaining words in 'CarName', assumi
cars['name'] = cars.apply(lambda x: ' '.join(x.CarName.split(' ')[1:]), axis=1)
```

In [ ]:
```
# Display all unique values in the 'brand' column to verify the extraction proce
cars.brand.unique()
```

In [ ]:
```
# Correct common misspellings and inconsistencies in car brand names within the
cars.CarName.replace({'maxda':'mazda', 'porcshce':'porsche','toyouta':'toyota',
                      'vw':'volkswagen', 'vokswagen':'volkswagen', 'Nissan':'nis
                     inplace=True)
```

In [ ]:
```
# Reconstruct the 'CarName' column to combine the corrected 'brand' and 'name' c
```

```
cars.CarName = cars.brand + ' ' + cars.name
```

In [ ]:
```
# Display the first few rows of the DataFrame to review changes made to the 'Car
cars.head()
```

Rename and drop columns

In [ ]:
```
# Rename specific columns for clarity and consistency within the DataFrame.
cars.rename(columns={'CarName':'brandname', 'car_ID':'carid'}, inplace=True)
```

In [8]:
```
# Display the first few rows of the DataFrame to review the renamed columns.
cars.head()
```

In [ ]: